

Customer Churn & Retention Analysis

IBM Telco Customer Churn dataset | Customer Analytics

Objective

Identify customer characteristics associated with churn and translate the findings into practical retention priorities.

Dataset and focus

The dataset contains 7,043 customer records. The analysis focuses on contract type, payment method, tenure and monthly charges.

Methods

Descriptive comparisons, Chi-square testing, odds-ratio reasoning and business interpretation.

Key retention signals

Contract structure and customer tenure are useful segmentation dimensions for retention activity. Payment method and monthly charges provide additional signals for prioritising customer outreach. These are associations rather than causal effects.

Business interpretation

Retention activity can be prioritised around customers in higher-risk contract segments and customers early in their tenure. Payment method and monthly charges can be used as additional segmentation variables rather than as standalone explanations of churn.

Portfolio value

Demonstrates structured data analysis, categorical hypothesis testing, interpretation of risk differences and translation of quantitative findings into business recommendations.

Reproducibility

The accompanying analysis.R file documents the analytical workflow and the repository contains supporting visualisations.